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# Towards Real-Time Full-Waveform LiDAR Transformers via Intensity-Guided Token Reduction and Physics-Aware Augmentation

Intensity-guided token prune/merge + waveform-safe aug.: first real-time FWLT

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Project page: keio-csg.github.io/FWL-ToPM (code and OOD dataset released)

TL;DR

# Summary: near-all-zero background buys both speed and accuracy

**FWL-ToPM**: prunes 70% of tokens by per-patch max intensity, orders the rest along a Morton curve and merges by similarity; **FWLAug**: mixing augmentation that copies whole waveforms and shifts along rays, preserving ToF structure and frustum geometry.

## 8×

### Speedup

Prior SOTA 6.54 → 53.2 FPS

Exceeds 10 Hz real-time requirement

## 0.592

### F1-mean

Prior SOTA 0.503 → +8.9 pts

±0.005 over 3 seeds

## 70%

### Tokens pruned

Accuracy rises, not falls, after pruning

0.555 → 0.591

## +4.6

### OOD F1 (pts)

FWLAug： 0.406 → 0.452

OOD Glass F1 +8.4 pts

# Background: richer FWL, but slow Transformers and scarce data

### Conventional LiDAR: peaks only → point cloud (x, y, z)

Keeps only a few high-intensity peaks per pulse

Glass multipath, fog/rain backscatter → ghost and noise points

Degrades detection and SLAM, posing safety risks

### Full-waveform LiDAR (FWL): full histogram (u × v × t)

Records each pixel's complete ToF intensity waveform

Retains multipath echoes and weak signals

Enables ghost detection, in-fog point cloud reconstruction, etc., beyond standard LiDAR

Task: per-voxel classification into Noise / Object / Glass / Ghost

## Two bottlenecks of FWL Transformers

### ① Not real-time

Far more data than point clouds, mostly low-intensity background

Neural DSP ≈ 2 FPS; FWL-MAE ≈ 6.5 FPS, both below 10 Hz

Image token reduction assumes information is uniform in 2D, wrongly dropping sparse high-intensity tokens

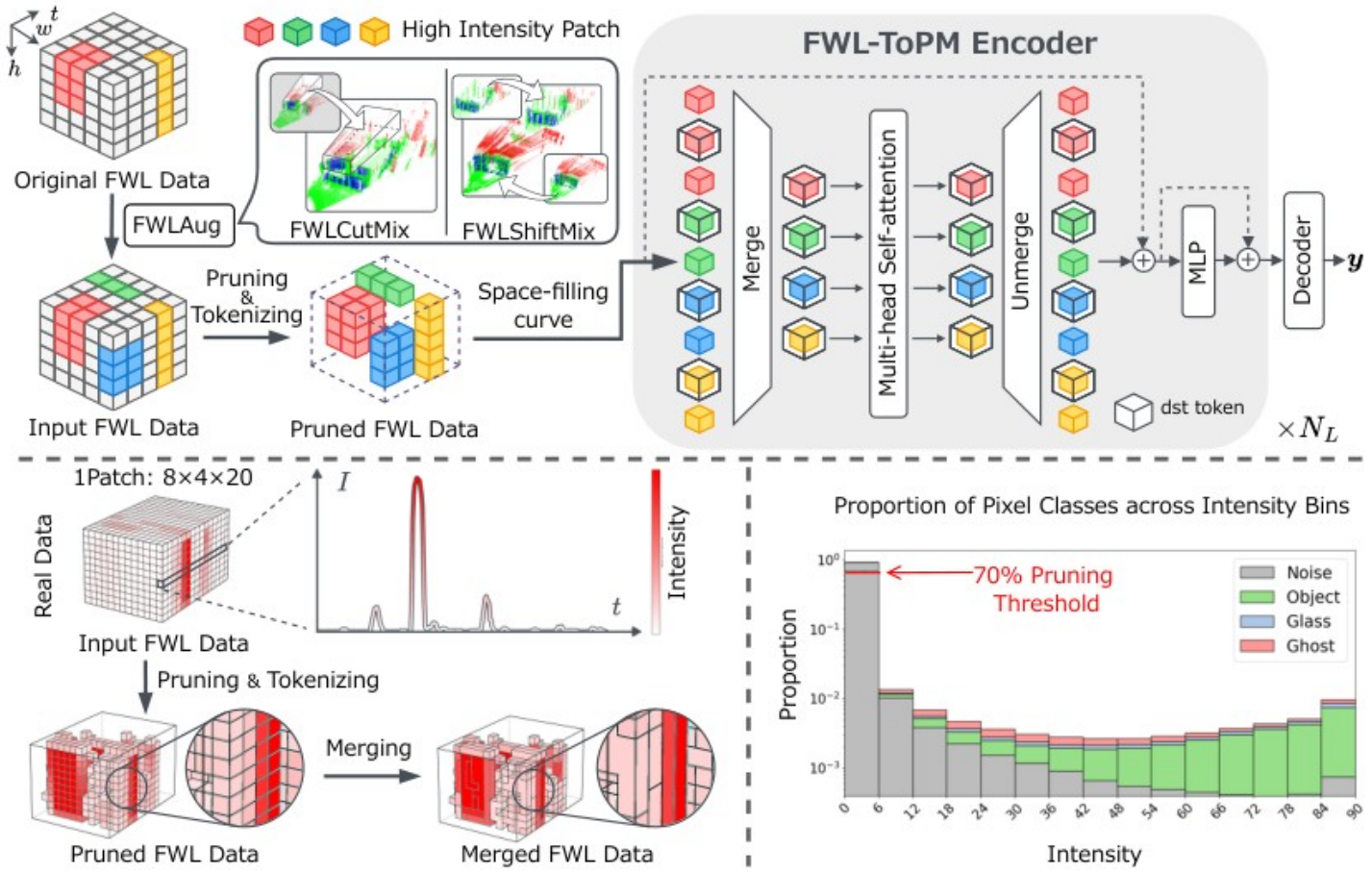
### ② Poor generalization

Ghost-FWL is the only public FWL dataset with peak-level labels

Minority classes (e.g., glass) are scarce; standard augmentation breaks waveforms and frustum geometry

**This work: FWL-ToPM solves ①, FWLAug solves ②**

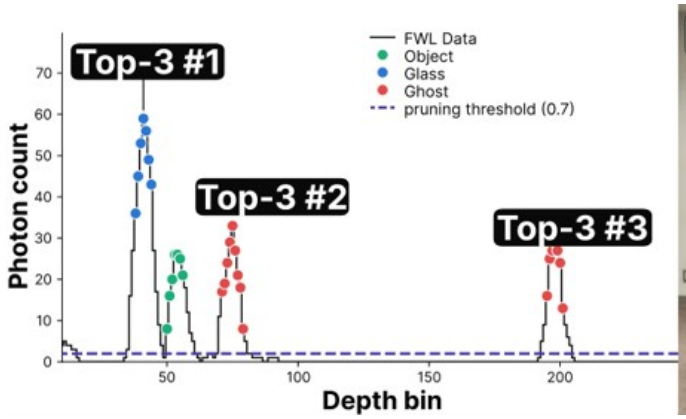
# Insight: most patches are weak noise; focus on rare bright ones

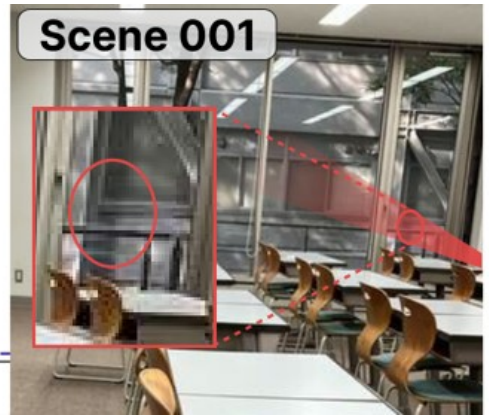


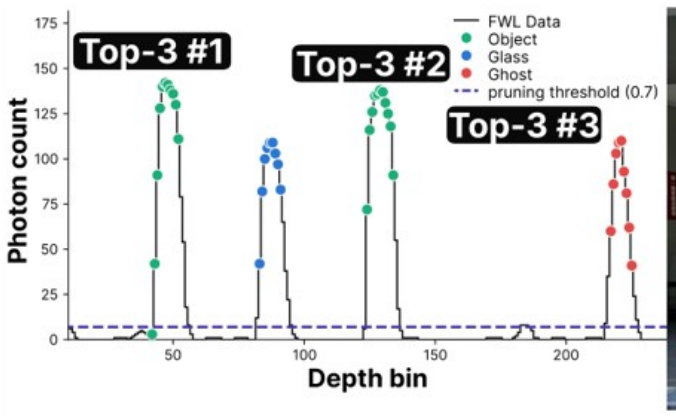

1. **FWLAug (training only)**
   Waveform-preserving mixing augmentation
2. **Intensity pruning**
   Drop the lowest 70% of patches
3. **Morton serialization**
   Preserves 3D adjacency
4. **Merge → Attention → Unmerge**
   Merging within each block, ρ_m = 0.9
5. **Linear decoder**
   Per-voxel 4-class logits

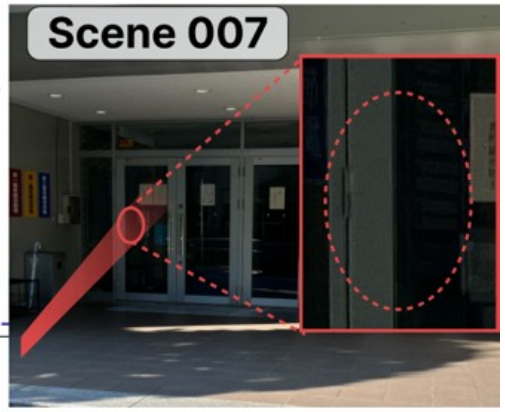
ViT: 6 layers / 6 heads / E = 240; patch = 8×4×20; focal loss

# Intensity pruning: drop 70% weak patches, safer than Top-K peaks




## Rule

m_i = max intensity within patch

Sort ascending; lowest ρ_low = 0.7 of patches skip the encoder

## Train / Inference

Training: no loss on pruned regions

Inference: pruned regions labeled Noise

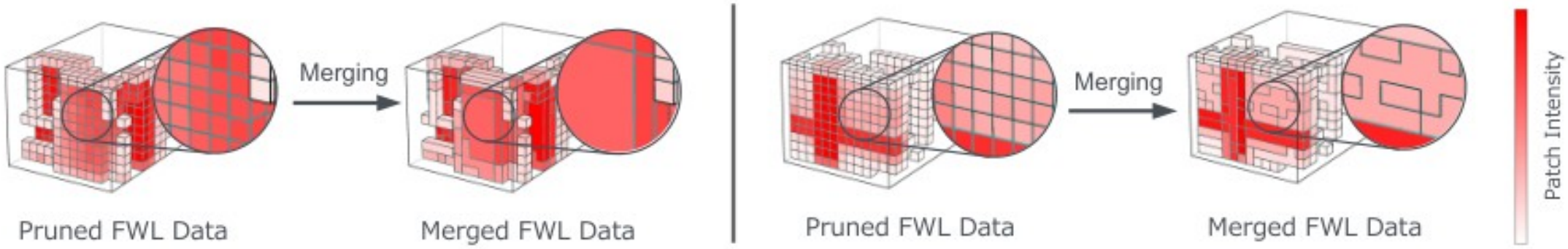
No network weights modified

## vs. Top-K peaks

A fixed per-ray peak budget lets strong ghosts crowd out Object / Glass echoes

Very low pruning threshold → these echoes kept with full waveforms

# Sparsity-aware merging: Morton order, uniform dst, similarity merge




**Morton serialization**
Pre-pruning 3D coords into a 1D sequence

**Evenly spaced dst**
$\rho_d = 0.1$
Spatially uniform

**Similarity merge**
Cosine-similarity assignment
Averaged at $\rho_m = 0.9$

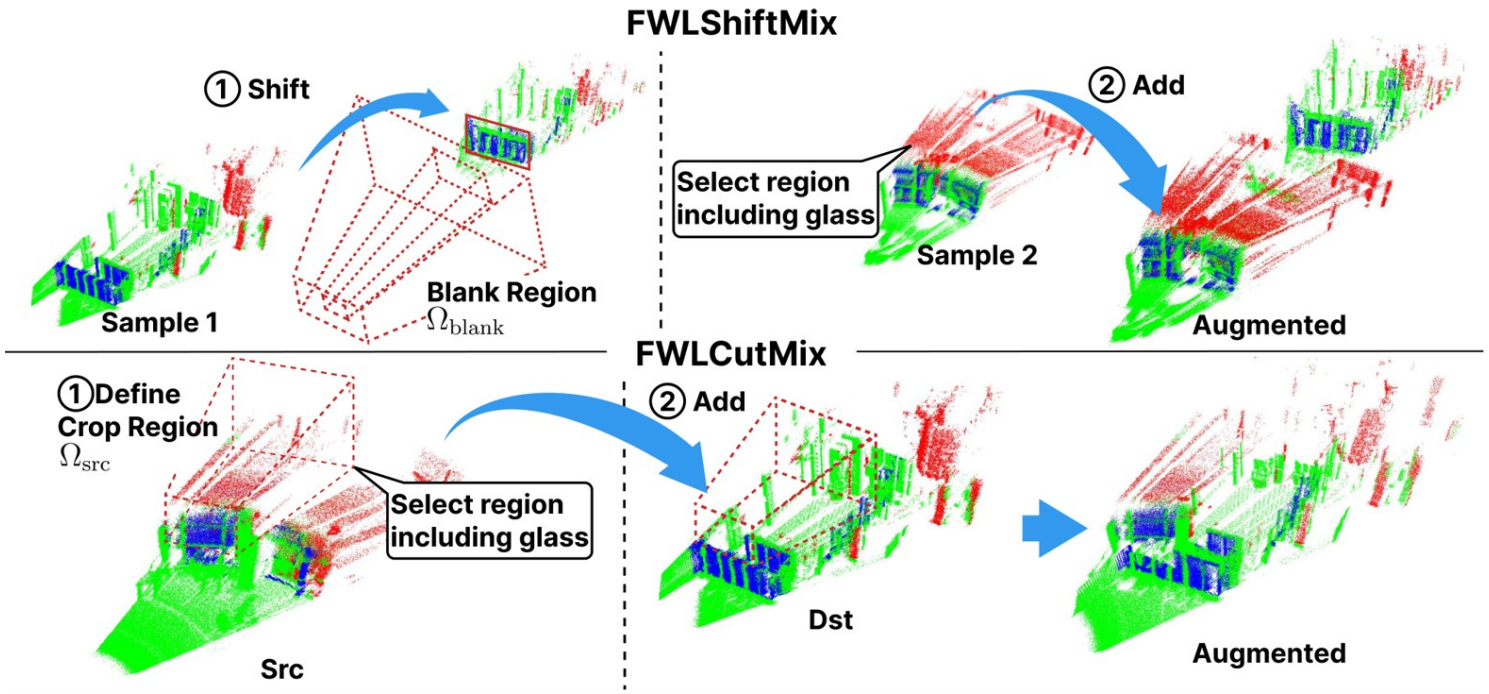
**Unmerge**
After attention, dst features copied back in place

Merging affects only attention; weights unchanged, so it applies directly to pretrained models

**Space-filling curve ablation (F1-mean)**

| Setting | Hilbert | Morton (ours) |
|---|---|---|
| Merge only | 0.564 | 0.565 |
| Prune + Merge | 0.589 | 0.592 |

# FWLAug: waveform copy + ray shift keep ToF/frustum geometry




## FWLShiftMix: range diversity

Peaks → 3D points, shifted along rays by Δd ~ U(−3, 15) m; waveforms shifted along T accordingly

Gaps left by moved objects are filled not with zeros but with whole waveforms from glass regions of another frame

## FWLCutMix: minority classes

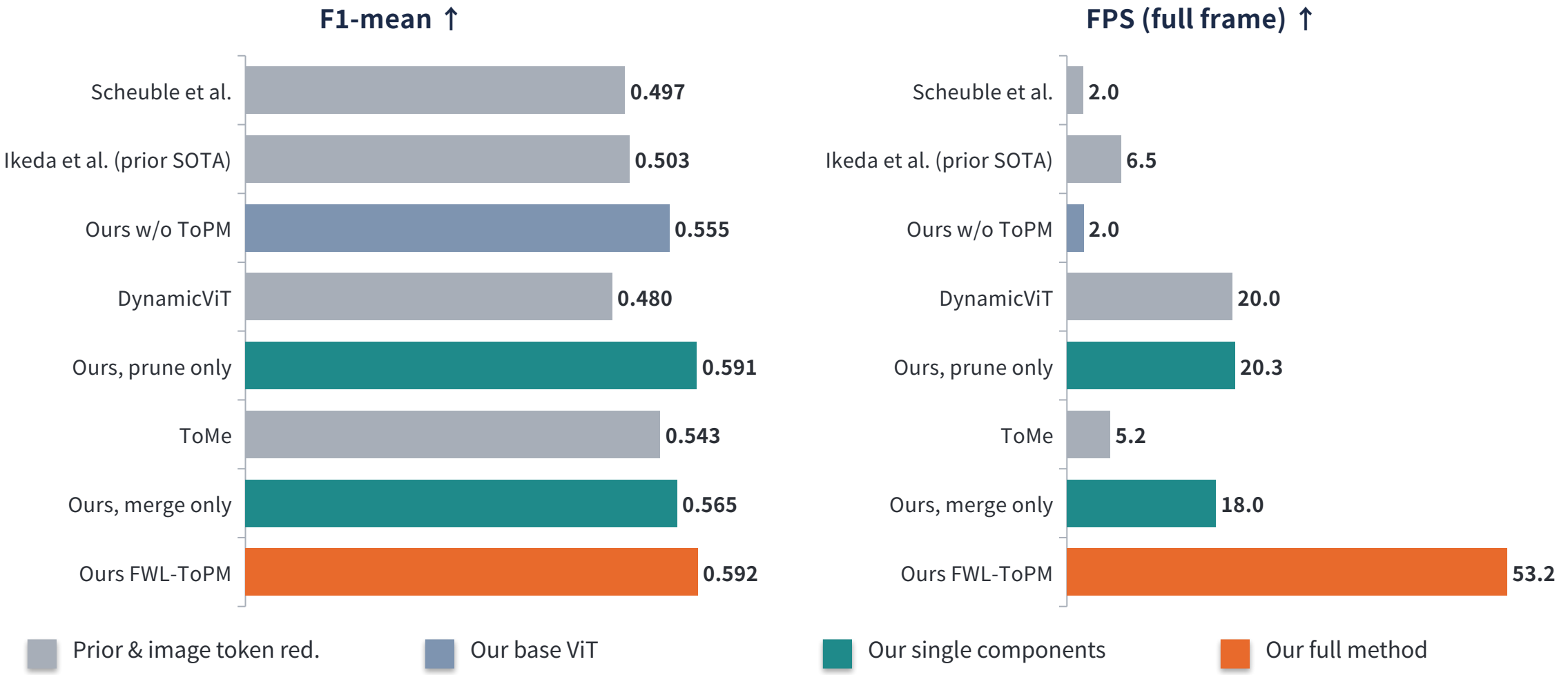
Crop size set by mix ratio λ; glass regions preferred

Full T-axis waveforms pasted at random target-frame locations (1–3 per step)

## Role: regularization, not optical simulation

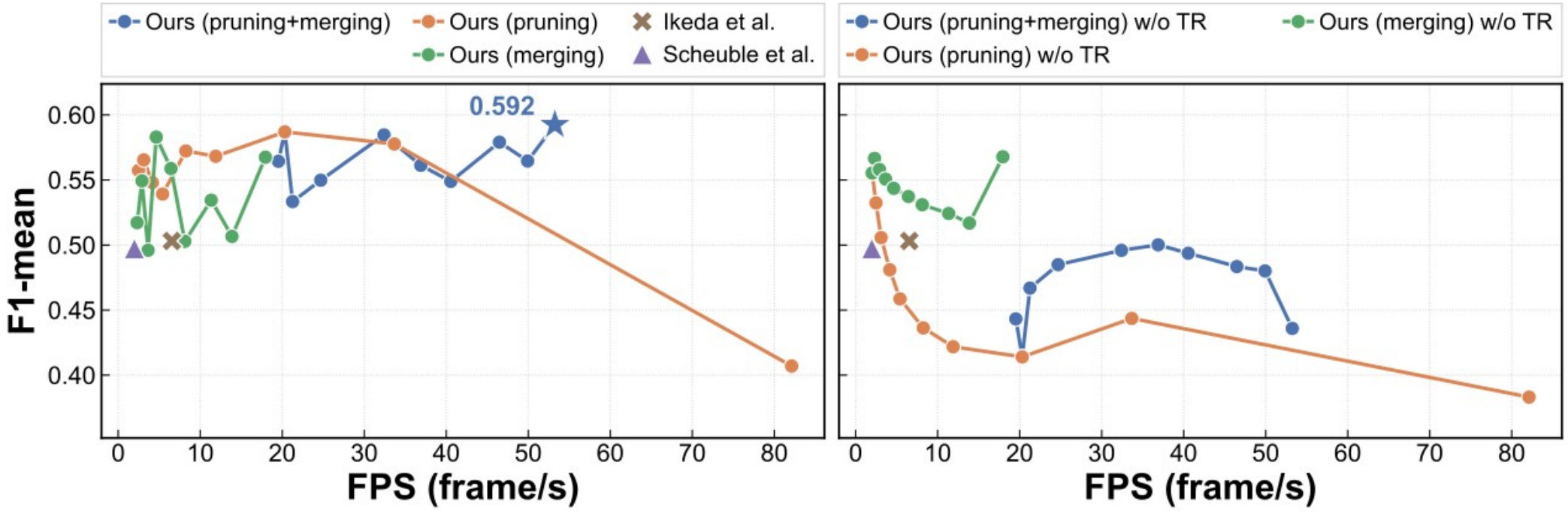
One of the two chosen per iteration (p = 0.5)

# FWL-ToPM achieves top F1 (0.592) at 53.2 FPS, 8× faster than prior SOTA



RESULTS • TRADE-OFF

# Complementary: prune-only degrades >20 FPS; merge-only caps ~18 FPS




**Prune only**

Peaks at 0.587 at ~20 FPS; drops to ~0.41 at ~82 FPS

**Merge only**

Max ~18 FPS; F1 fluctuates in 0.50–0.58

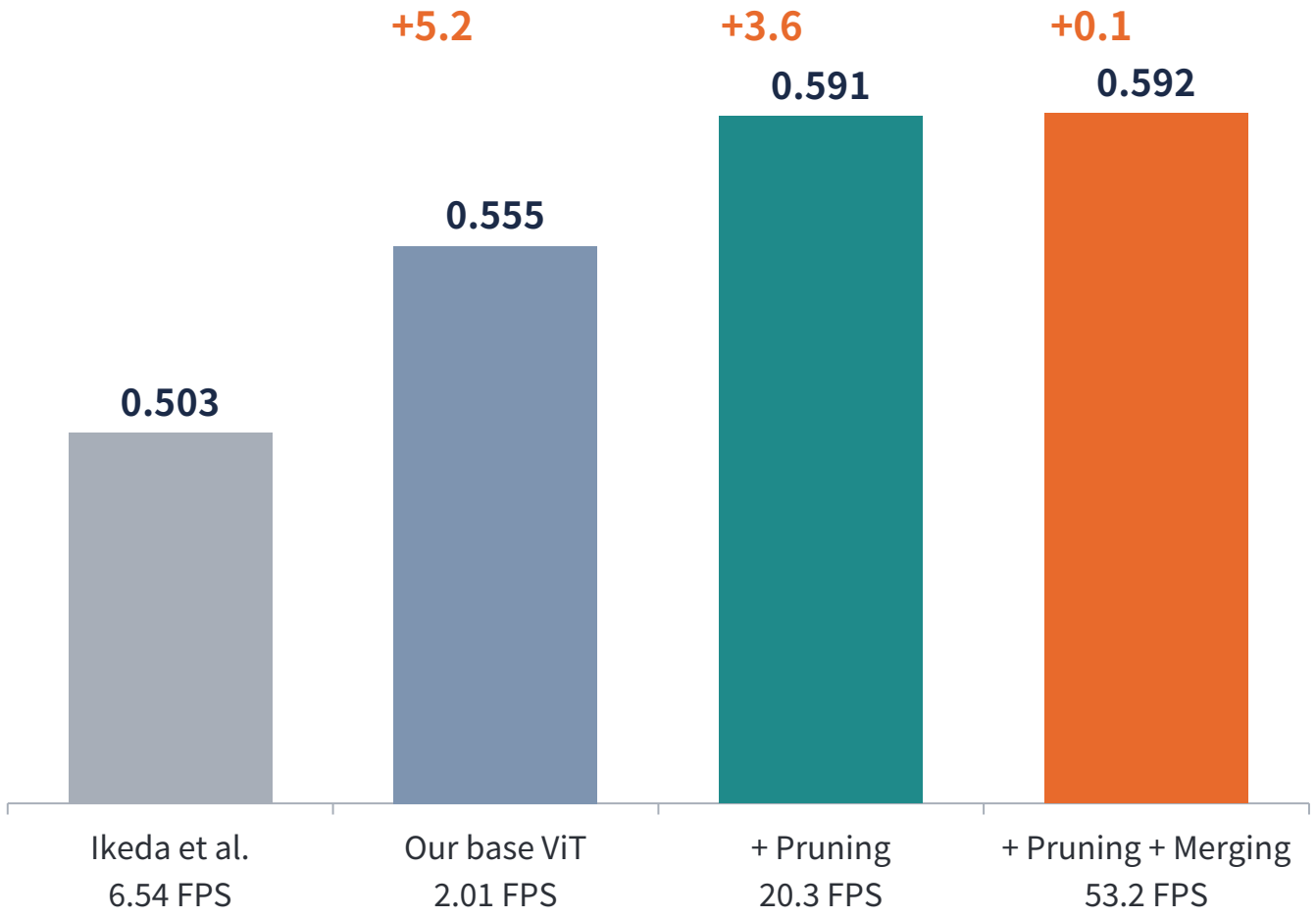
**Prune + Merge**

~0.53–0.59 over 20–53 FPS; 0.592 at 53.2 FPS

**Training-free (right)**

Pruning quickly collapses to ~0.41–0.44; merging keeps 0.52–0.57 at ~18 FPS

# +8.9 pts: base ViT 5.2, pruning 3.6; merging mostly adds speed




Note: y-axis starts at 0.40 to magnify differences

**Per-class F1 (mean of 3 seeds)**

| Class | Ikeda et al. | Ours |
|---|---|---|
| Object | 0.680 | 0.741 |
| Glass | 0.268 | 0.358 |
| Ghost | 0.561 | 0.678 |

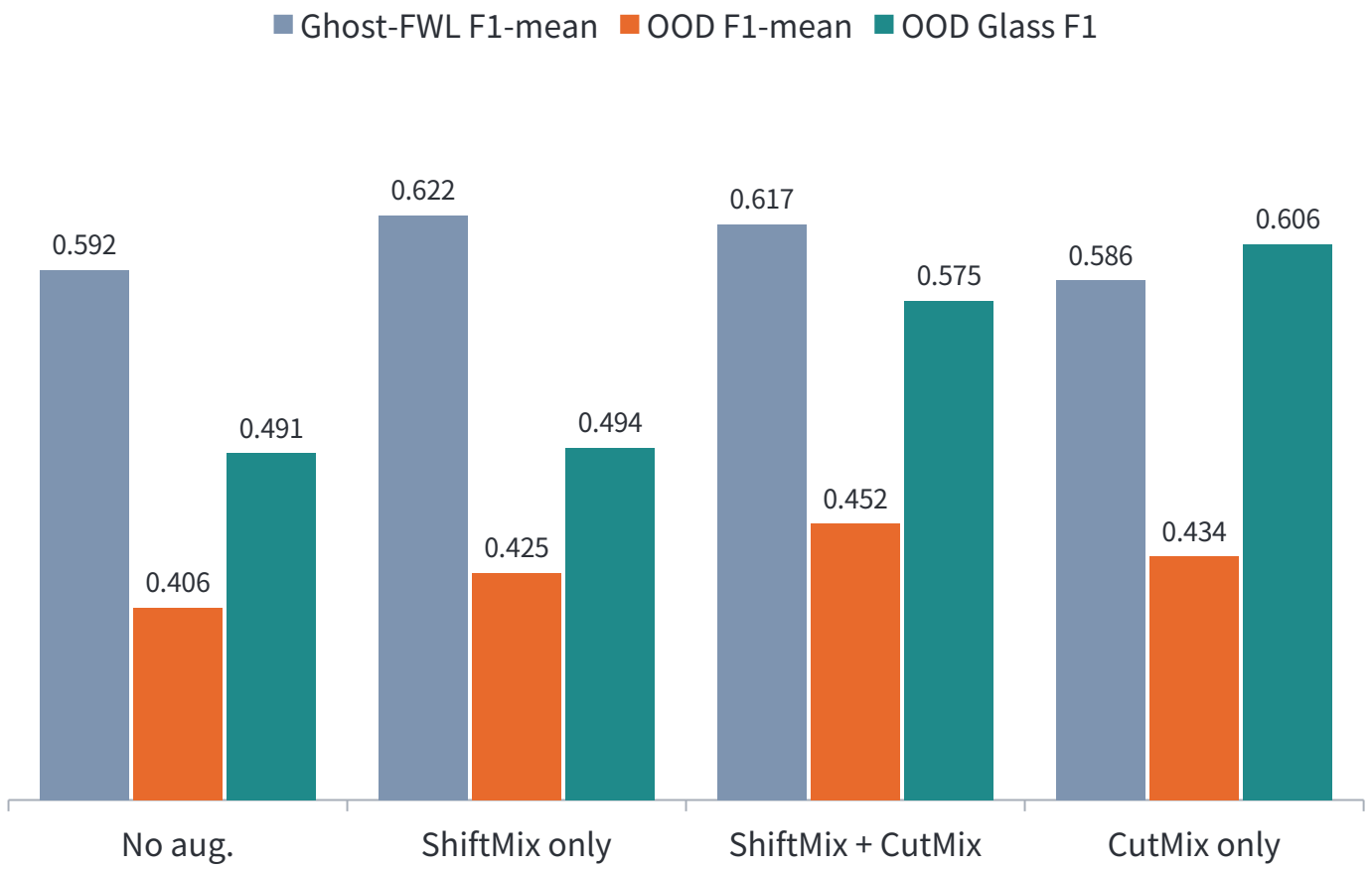
## Interpretation

Base ViT with 8×4×20 3D patches and other designs already beats prior SOTA

Accuracy gain comes almost entirely from pruning; merging takes 20 → 53 FPS

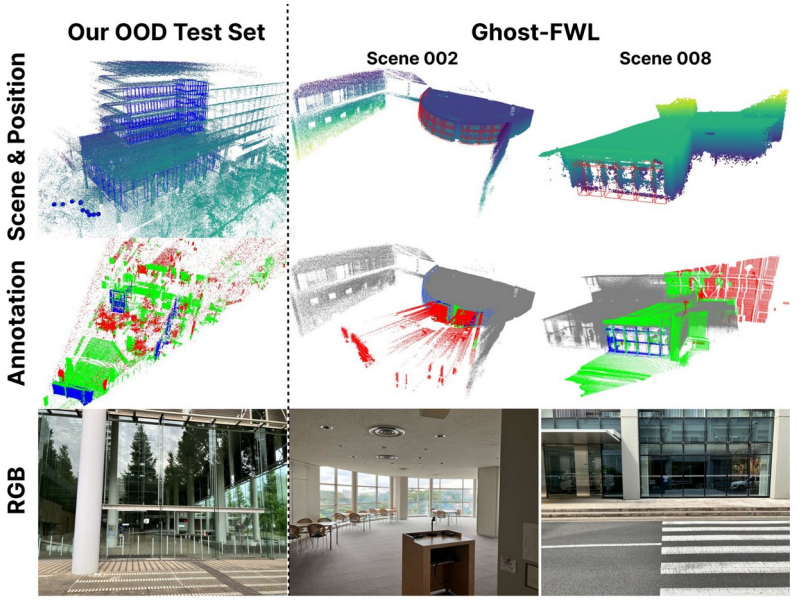
~26× faster than our own base model (2.01 → 53.2 FPS)

# FWLAug aids OOD generalization: F1 +4.6 pts, Glass F1 +8.4 pts




**ShiftMix** mainly boosts in-domain (0.622); **CutMix** mainly boosts OOD Glass (0.606); combined gives best OOD F1 (0.452)

Note: y-axis starts at 0.30

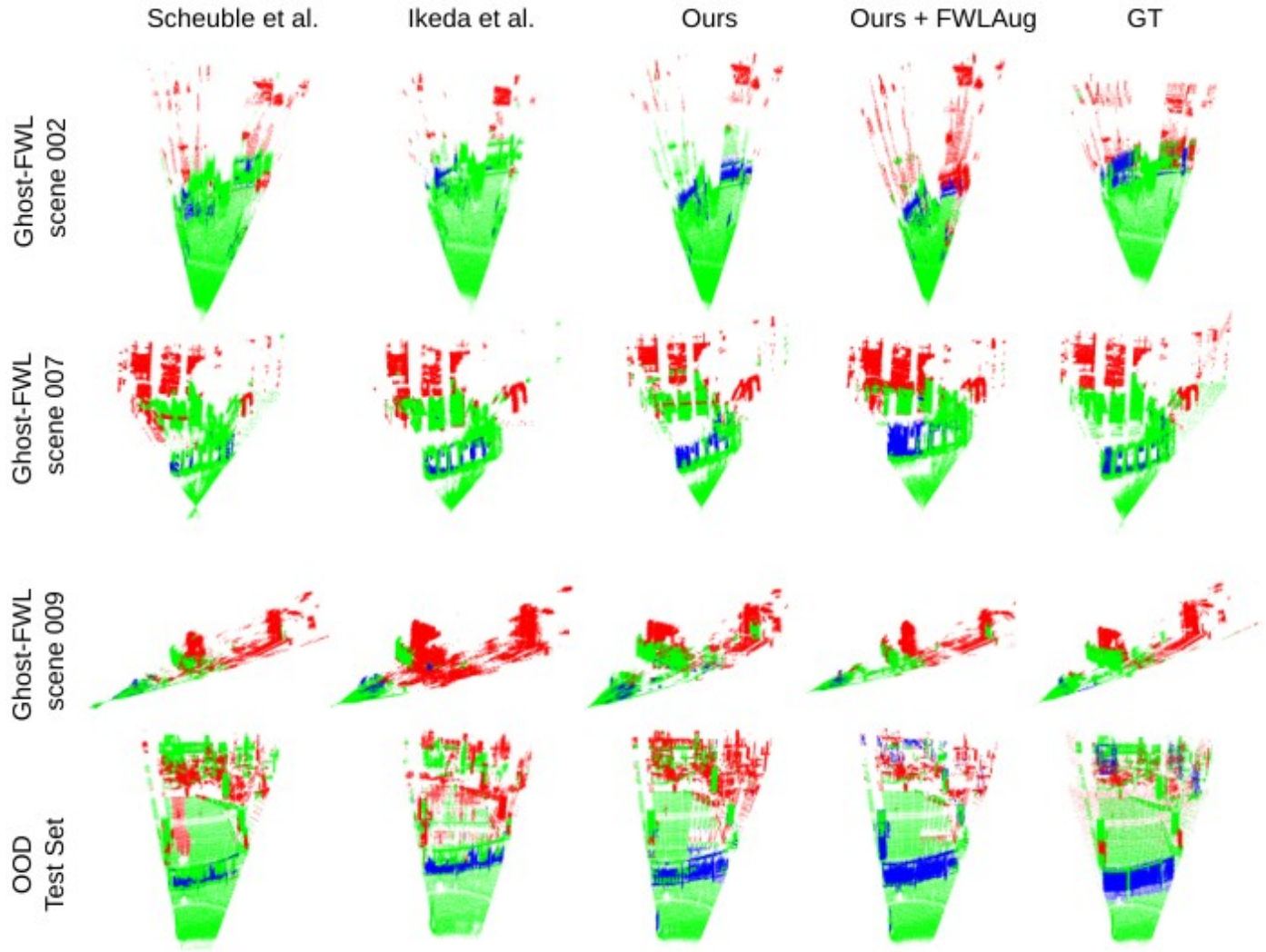



## OOD test set (self-collected, public)

500 frames / 10 viewpoints; 7-story building glazed on all four sides, 300+ glass panes

Same sensor, settings and labeling pipeline as Ghost-FWL → only the scene distribution changes

# Qualitative: glass/ghost closer to GT; FWLAug fuller in glass-dense scenes




## Colors

Ghost  Object  Glass

## Improvements

Ghost-FWL scenes: more accurate Glass / Ghost

OOD scenes: better classification of objects behind glass; FWLAug yields more complete glass regions

## Failure cases

"Glass behind glass" remains hard

FWLAug detects it only partially, with errors

Some glass misclassified as objects

# 6 caveats for interpretation

## 01 Source of gains

Of +8.9 pts, 5.2 come from base ViT design; FWL-ToPM adds 3.7, almost all from pruning

## 02 Speed metric

Model-level throughput; full frame = 4 sliding windows in sequence; excludes sensor I/O, preprocessing, embedded deployment

## 03 Curve noise

Adjacent configs in Fig. 3 differ by up to ±0.03; 3-seed std reported only for final config

## 04 Strong pruning prior

Pruned regions are set to Noise, which F1-mean excludes; a fixed 70% may prune distant / low-reflectance / in-fog weak signals

## 05 Training-free & fairness

Training-free holds only for merging; DynamicViT prunes only at input, ToMe merge ratio capped at 0.5 (ours 0.9)

## 06 Limited OOD evidence

One sensor, one OOD site; full FWLAug not best on every metric; trained on only 1/3 of Ghost-FWL

TAKEAWAYS

# Takeaway: intrinsic intensity sparsity → first real-time FWL Transformer

### Real-time

**53.2 FPS • F1 0.592**

Intensity pruning + Morton-order merging: first real-time FWLT (8× faster than prior SOTA, +8.9 pts)

### Less is more

**Prune 70% → acc. ↑**

Low-intensity tokens dilute object signals like noise; pruning them: 0.555 → 0.591

### Generalization

**OOD F1 +4.6 pts**

Whole-waveform copy / along-ray shift augmentation: OOD Glass F1 +8.4 pts in glass-dense unseen scenes

### Toward deployment

Adaptive ρ_low / ρ_m / ρ_d to protect weak signals • Validation across sensors, resolutions and adverse weather • Sensor-to-decision system latency and embedded deployment